IT2140 - Database Design and Development

Year 2, Semester 1

Tutorial 02

Learning outcomes:

- Create relational tables with appropriate constraints (e.g., NOT NULL, CHECK)
- Perform data manipulation using INSERT, UPDATE, and DELETE statements
- Modify table structures and enforce referential integrity using foreign key constraints
- Write SELECT queries to retrieve, filter, sort, from tables

Employee (eid, ename , salary, dept, phone)

Department (deptNo, dname, location, mgrID)

1. Create the *Employee* table *Department* tables with following constraints

    a. Employees' name cannot be null

    b. Employees' salary should be greater than zero

    c. Department location can be 'Colombo', 'Kandy' or 'Galle'

    ** Do not add the foreign key constraints to the table at this point

2. Insert following rows to the *Employee* table

eid	ename	salary	dept	phone
E0001	Kumari	45000	3	0112123456
E0002	Anushka	63000	1	0112123457
E0003	Anura	27000	2	0112123458
E0004	Niranjala	36000	3	0112123459
E0005	Sampath	50000	1	0112123450

3. Insert the following rows to the *Department* table

deptNo	**dname**	**location**	**mgrID**
1	Administration	Colombo	E0002
2	Sales	Kandy	E0002
3	Finance	Colombo	E0005

4. Add the foreign key constraints to the created tables.
5. Add a column named *bdate* to the employee table to store the birth dates of employees.
6. Update the salary of Kumari to 35000.
7. Delete Sampath from *Employee* table.
8. Display the employee names and their salaries sorted by their names.
9. Display the names of the employees who obtain a salary greater than 50,000.
10. Display the employee's name and the id of the department he/she is working in.